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QUALITY PLAN – EEG FIELD KIT

Document: QP-EEG-010 **Revision:** B **Date:** 2026-09-01 **Issued by:** TI One Voice research programme (one.witysk.org), Brussels, Belgium **Governing documents:** DSN-EEG-003 Rev D, then RFQ-EEG-001 Rev E. Where this document and design.py disagree, design.py governs. **Revision note (Rev B):** carrier corrected to 150.0 x 130.0 mm and to four layers, with inner-layer registration, plane continuity and misregistration added to incoming inspection, first article and the delivery audit; through-hole part and hole counts recounted from design.py; the T7 10 uV limit and its k = 2 uncertainty corrected; the C1-C16 and C21/C41/C61 part numbers, the fiducials, the conformal-coating decision, the serial format, the per-unit isolation test and the headphone load brought into line with the package rulings. The findings of the second cross-document audit of 1 September 2026 are closed in this issue: the "has passed the DRC" claim is replaced by the measured routing result and the twenty-five open DRC items, the layout-rule citation is corrected to DSN-EEG-002 section 13, the governing-document revisions are restated, the rulings are cited as the controlled document RUL-EEG-021, and incoming-inspection row IQC-B15 is added so that a fabrication release is accepted only against a DRC report with zero open items. The revision letter is unchanged: these are corrections within the same release.

Further corrections (2026-09-01, third audit), revision letter again unchanged: E-11 is no longer written as though TST-EEG-004 had widened a limit to accommodate the X7R capacitors – both Sallen-Key capacitors are X7R at 15 %, f0 moves to 42.4 to 57.4 Hz, that is wider than E-11's 45 to 55 Hz, and the incoming row and FA-09 now say the 50 Hz half of E-11 is not met and that T12e's 42 to 58 Hz is the band this package sets; **E-23 is stated as met in part everywhere it appears**, with the thermal-regulation half met on the charger IC and the 45 C inhibit half not met and untestable for want of a cell temperature; and **the pad census in section 1 says what it counts**, against AVL-EEG-017's smaller figure. The Sallen-Key capacitors are also corrected from 50 V to the 25 V `design.py` actually fits.

The routing correction (2026-09-02), revision letter again unchanged: the Rev B DRC report now records **zero violations** – all 145 nets fully connected, no net without copper, both inner planes continuous under the analogue zone – so the twenty-five open DRC items stated throughout the previous issue of this plan are gone, and so are the two 0.328 mm electrode-clearance vias. Section 1.1, row IQC-B15, row FB-09 and open item 13 of section 14.2 are restated accordingly: the fabrication data is **released for review under RFQ-EEG-002A** and is **not released for fabrication**, because no human layout engineer has looked at routing produced by the programme's own tools and 169 of its connections close at the minimum conductor or the minimum gap rather than the preferred width. No hardware in this package has been built or measured. S-04, with E-23's 45 C charge inhibit, is untouched by the routing and stays not met. *Corrected 2026-09-02: this sentence also named "S-02 at 53.2 uA" as staying not met. ECO-EEG-024 was applied the same day and S-02 is met in the design at 36.8 uA; see the correction below.*

Corrections after the firmware build and ECO-EEG-024 (2026-09-02), revision letter again unchanged. Two things this plan asserted throughout stopped being true on the day it was issued, and both are restated rather than deleted. **S-02 is met in the design.**

ECO-EEG-024 is applied in `tools/design.py`: R1 to R16 are **68 kOhm** (Vishay TNPW060368K0BEEA), the single-fault DC current is **36.8 uA** against the 50 uA limit, and E-10 moves to the **+/- 1.0 dB** branch the requirement already carried. Row IQC-2.3, row FA-03, section 1.4's pointer row and open item 3 of section 14.2 are restated; the 47 kOhm figures are kept beside the new ones as the superseded set. **Met in the design is not signed off**: nothing is measured, no unit exists, and the electrical safety reviewer named in open item 1 has not started, so S-02 remains that reviewer's item. **The contact-light driver is written.** `firmware/main/main.c` implements E-27's bicolour phase scheme – both halves of the converter's lead-off status are captured, neither detector gone is green, exactly one is amber, both is red – so open item 7's "the bicolour phase scheme is specified and not yet coded, so the contact-light test step cannot pass" is superseded. TST-EEG-004 T11 is written against it and has never been run: no unit exists. The firmware itself is now **built** – ESP-IDF v5.2.5, target esp32s3, images and SHA-256 manifest in `firmware/release/` – and has run only under QEMU emulation, never on hardware.

Corrections after the verification review of package v2.2 (2026-09-02), revision letter again unchanged: two figures are corrected to the artefacts they describe. Row IQC-B15 dated the released DRC report 1 September; `kicad/EEG-CAR-01_RevB_DRC_report.txt` is dated 2 September on its own `Generated` line, and the date an inspector compares is the date printed on the report. Section 12.4's fixture self-test named the lead-off references by the rounded values 5 k / 10 k / 50 k; FIX-01/A switches the E96 parts **4k99 / 10k0 / 49k9**, which is what JIG-EEG-009 Rev C section 0.1 and TST-EEG-004 Rev C T10 already name, and the rounding is not harmless at the top of the range. Both are corrections of wording to fitted fact; nothing about the plan, the sampling or the limits changes.

Why this document exists

The v1 package named quality in three fragments – "visual and AOI inspection to IPC-A-610 class 2" in RFQ-EEG-001 section 9.1 and TST-EEG-004 step T1, "IPC class 2" on the PCB specification sheet, and "X-ray on the converter packages for the Phase 1 units" – and nothing tied them together. The 14-agent audit of v1 raised five findings in this domain: `quality-plan-aql` (no inspection points, no sampling, no first-article procedure, no non-conformance route, no ESD requirement), `iqc-incoming-inspection-modules` (no incoming inspection for the twelve purchased module types the whole architecture rests on), `sampling-plan-and-msa` (no sampling plan, no gauge R&R, no golden unit, no fixture verification interval), `failure-disposition-and-rework` (nothing says what happens when a unit fails), and `supplier-quality-and-avl` (no supplier quality requirements or change notification). Contradiction XD-15 additionally showed that a review finding recorded as closed – the correction of a bidder's certification status – had never been applied. This plan closes all six. It is a contractual annex to RFQ-EEG-001 Rev E: bidders price against it, and the programme audits against it.

No hardware in this package has been manufactured or measured; the firmware image in `firmware/release/` is the only thing that has been built, and it has run only under emulation. **No safety engineer has reviewed this design.** Every figure below marked *calculated* is exactly that, and every limit that depends on a measurement the programme has not yet made is flagged as such.

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1. Scope, and the standards actually claimed

This plan covers the carrier board EEG-CAR-01 Rev B (150.0 x 130.0 mm, **four layers**, 211 reference designators, **636 pads in total on the board**, 156 nets, 30 connectors J1 to J30 of which seventeen are module connectors), the twelve purchased module types, the helmet and harness assemblies, the kit contents and the packed travel case, across all three phases: Phase 1 (2 prototypes), Phase 2 (10 kits) and Phase 3 (10 to 40 further kits, 25 to 50 in total).

Of the 211 designators, 186 are purchased placements: 153 surface-mount, of which **R89 is do-not-populate**, and **33 through-hole**. The remaining 25 designators are 18 test pads, 3 fiducials and 4 mounting holes. Thirteen module assemblies are fitted per unit from the twelve purchased types, because the ADS1299 breakout is fitted twice. Twelve assemblies mount on the MP-01 module plate; the ESP32-S3-DevKitC-1-N16R8 is inserted directly into J6 and J7.

What the 636 counts, exactly. Every pad in `tools/design.py`, with nothing left out: 390 surface-mount, of which 18 are test pads and 6 are the pads of the three fiducials; 236 plated through-hole; and 10 non-plated, being the four mounting holes and the six anchor holes of the DIN sockets J15 to J17. **620 of the 636 carry a net**, so the netlist pin count is 620, not 636; the 16 that carry none are the 6 fiducial pads and the 10 non-plated holes. This plan inspects against the 636, because a stencil, an AOI recipe and a bare-board test all have to account for pads that no netlist mentions.

AVL-EEG-017 Rev C section 1 gives 614 for the same board, and that figure is not this one. The difference is exactly 22: the 18 test pads plus the 4 mounting-hole pads. So 614 is this board with those left in place and left out of the count – the six fiducial pads are in both figures – and it is neither a different board nor the 620 netlist pins. AVL-EEG-017 does not say which pads its figure omits. Where a pad count is inspected against, this plan's 636 governs, and AVL-EEG-017 is corrected to it at its next revision.

1.1 Two things changed during layout, and this plan inspects both

The board grew from 130 x 124 mm to 150 x 130 mm. Thirty connectors, 211 parts and 156 nets would not close at the smaller size. The extra 33.8 cm^2 of bare board costs a few euro per unit at these quantities, against a real risk of an unroutable design.

The board went from two layers to four. Package v1 asserted that a two-layer carrier was enough. Doing the layout showed that it is not: on two layers the bottom side has to be both the reference plane and the second routing surface, and it cannot be both. Four layers – L1 signal, L2 reference plane, L3 reference plane, L4 signal – give two full routing surfaces and a continuous reference under every analogue trace, which is what DSN-EEG-002 section 13's "layout rules that are requirements, not preferences" require, and which a swiss-cheesed two-layer pour cannot deliver. At 2 units the four-layer board costs about EUR 35 more in total; at 50 units it is about EUR 3 per board. For a sixteen-channel EEG front end that is the right trade, and it is the single most important thing package v2 learned by doing the work instead of asserting it.

Two citations in that sentence are worth being exact about, because this plan inspects against both. The layout rules "that are requirements, not preferences" are **DSN-EEG-002 section 13**. The zoning rule, the star-point rule and the isolation keep-out are **DSN-EEG-003 section 3.3**. DSN-EEG-003 runs to section 11 plus its annexes and has no section 13, so a citation to "DSN-EEG-003 section 13" is wrong wherever it appears.

The quality consequence is that three characteristics which did not exist on a two-layer board now have to be inspected on every fabrication lot: **inner-layer registration**, **plane continuity on both inner layers**, and the **layer-to-layer misregistration limit**. They are rows IQC-B11, IQC-B12 and IQC-B13 in section 2.1, rows FB-21 to FB-24 in section 4.3, and row A-23 in section 13.

The geometry this plan inspects against, taken from `design.py` and `mech_gen.py`:

Item	Value
Carrier outline	150.0 x 130.0 mm, rectangular, no cut-outs
Layers	four: L1 signal, L2 reference plane, L3 reference plane, L4 signal
Stack-up	mask / 35 um L1 / prepreg 0.200 / 17 um L2 / core 1.065 / 17 um L3 / prepreg 0.200 / 35 um L4 / mask = 1.60 mm +/- 10 %
Copper	1 oz (35 um) outer, 0.5 oz (17 um) inner, both finished
Reference planes	AGND_REF left of x = 62 mm and DGND right of it, on **both** L2 and L3, tied together by stitching vias
Vias	**through vias only** – no blind, buried, back-drilled, filled or plugged vias. 0.60 mm pad on a 0.30 mm finished hole, 0.15 mm nominal annular ring, tented both sides
Zone split	x = 62 mm: analogue left, digital right
Mounting holes	M3, 3.2 mm NPTH at (5,5), (145,5), (5,125), (145,125), 6 mm copper keep-out
Isolation keep-out	x >= 141 mm, y = 2 to 22 mm, no copper on any of the four layers
Fiducials	three 1.0 mm round fiducials with 3.0 mm mask openings, at (12,10), (144,100) and (12,120)
MP-01 module plate	146.0 x 126.0 x 3.0 mm, 8 mm solid border, 12 x 3 mm jumper slots on a 16 x 7 mm grid
POD-P1 base	163.0 x 143.0 x 58.0 mm external; 158.0 x 138.0 x 55.5 mm internal
POD-P1 lid	163.0 x 143.0 x 6.0 mm, 2.0 mm spigot
Stack budget	floor 2.5 + boss 6.0 + carrier 1.6 + standoff 18.0 + plate 3.0 + modules <= 18.0 = 49.1 mm against 55.5 mm internal, margin 6.4 mm

The Rev B routing, and what it does and does not permit. EEG-CAR-01 Rev B is routed on four layers with 3 745 track segments and 552 through vias, and each reference plane is one continuous island per net. **All 145 nets are fully connected**:

none unclosed, none without copper. Every rule in `kicad/EEG-CAR-01_RevB_DRC_report.txt` passes: the smallest measured clearance is 0.260 mm on L1, 0.285 mm on the planes and 0.275 mm on L4 against a 0.20 mm rule; the narrowest conductor is 0.20 mm and the smallest plated hole is 0.30 mm; no copper comes within 2.00 mm of a non-plated hole; there are no zone crossings, no duplicate copper segments and no duplicate via positions; no digital net enters the analogue zone; and there is exactly one AGND_REF-to-DGND bridge and one HARN_SHIELD-to-DGND bridge. The report's own line is "VIOLATIONS: 0 – none. The board passes every rule listed above." It also records the isolation strip as free of copper on all four layers. That last statement is made because the report says it, not because the keep-out was assumed to have been honoured on the inner layers.

The board closes, and the data is released for review, not for fabrication. ECO-EEG-016 section 3 gates a fabrication-data release on zero DRC violations, every net one connected copper island, and both inner planes continuous under the analogue zone. **All three are now met**, so the data is **RELEASED FOR REVIEW under RFQ-EEG-002A** – and **fabrication release awaits that review**, because the routing was produced by the programme's own tools and has **not** been reviewed by a human layout engineer. How the board closes is part of what that reviewer is for. **169 connections are relaxed**: 36 take a conductor narrower than the 0.25 mm preferred width, and 133 keep full width and take a reduced gap instead, every one of them at or above the 0.20 mm minimum conductor and the 0.20 mm minimum gap. A board that closes at minimum geometry is not the same board as one that closes at preferred geometry, even when every rule passes. Row IQC-B15 in section 2.1 remains the inspection point that keeps a board built to a data set that is released for review under RFQ-EEG-002A but not yet released for fabrication out of the build, and the four-layer change is itself a change the layout and safety reviewer must check when one is appointed.

RUL-EEG-021 section B gives the fiducial positions as (12,10), (144,100) and (12,120), which is where `design.py` places them and what is inspected. The register's first issue transcribed them as (8,8), (142,8) and (8,122); that error is corrected, and `design.py` governs in any case.

1.2 Standards required

Standard	Revision	Applies to	Required?
IPC-A-610 class 2	Rev H or later, stated in the quote	Assembled carrier workmanship, T1	**Required**
IPC J-STD-001 class 2	Rev H or later	Soldering process, SMT and the 236 plated through-holes	**Required**
IPC-A-600 class 2 / IPC-6012 class 2	current	Bare board acceptance, including the four-layer characteristics of section 2.1	**Required**
IPC-7711/7721	current	Rework and repair, with the limits in section 7	**Required**

Standard	Revision	Applies to	Required?
IPC-9252	current	100 % bare-board electrical test to the supplied IPC-D-356A netlist	**Required**
ANSI/ESD S20.20 or IEC 61340-5-1	current	An EPA covering bare carriers, ADS1299 modules, U1-U3 and U7	**Required**
ISO 9001	2015	Manufacturer's quality system	**Not required.** Asked for as information only
ISO 13485	–	Medical device quality system	**Not required, and not wanted**

All three IPC classes apply and all three are named: IPC-6012 class 2 for fabrication, IPC-A-600 class 2 for bare-board acceptance, IPC-A-610 class 2 for assembly. Where another document in the package lists only two of them, this row and DSN-EEG-003 section 3.2 govern.

1.3 Why ISO 13485 is not asked for, and a correction to the record

The instrument is a loaned research field kit. It is not a medical device and is not placed on the market. It is designed to the applicable principles of IEC 60601-1 for a type BF applied part (RFQ section 8) because that is good engineering for something connected to a person's head, but certification is not requested and no medical claim is made. Requiring ISO 13485 would add cost and lead time without adding assurance for this product, and it would give a false signal about the device's regulatory status. Bidders who hold ISO 13485 are not preferred over bidders who do not.

Correction to the record. DSN-EEG-003 Rev A.2 section 7, adversarial-review finding 11, recorded that "Regulus corrected our assumption of ISO 13485 – contact list was wrong for one company", with the action shown as complete. The correction was never applied to the file. Regulus Electronics (New Taipei, Taiwan) holds **ISO 9001 and does not hold ISO 13485**; the manufacturer contact list continued to assert "ISO 13485 capable" against them until package_v2.2. Two further contact corrections recorded as complete were also never applied: RayPCB should read PCBSync (contact stan@pcbsync.com), and NextPCB was to be dropped from the mailing but remained listed twice. All three are corrected in AVL-EEG-017 Rev C, which is now the single source for supplier identity and certification status. No certification claim about any bidder is to be restated in a programme document without a copy of the certificate on file.

1.4 What this plan does not restate

One table, one home. This plan cites the following and does not repeat them:

Content	Its one home
Board specification (size, stack, finish, tracks, holes, classes)	DSN-EEG-003 section 3.2

Content	Its one home
Isolation keep-out and the star-point rule	DSN-EEG-003 section 3.3
Module-to-connector table and the jumper set	ICD-EEG-006 section 1
ESP32-S3 GPIO map	ICD-EEG-006 section 5
ECO register	ECO-EEG-016 section 2
68 kOhm / 10 nF noise-and-flatness arithmetic (47 kOhm before ECO-EEG-024)	RISK-EEG-011 section 4, **which has not yet been restated against the applied ECO**
Test step numbers and their names	TST-EEG-004 Rev C
Public-key fingerprint definition	FW-EEG-001 section 7
Lithium shipping procedure	PKG-EEG-015 section 7
Foam pocket schedule and label artwork	PKG-EEG-015 sections 2 and 5
Serial format `TIOV-B-nnnn`	PKG-EEG-015 section 5
The rulings that settled the cross-document disagreements	RUL-EEG-021 Rev B

The geometry recap in section 1.1 is the one deliberate exception: the two layout changes must be visible in every document of the release, and the inspection rows below refer back to it rather than repeating the coordinates a second time.

1.5 Roles and sign-off authority

Role	Held by	Signs
Programme technical lead	TI One Voice, Brussels	ECOs, deviations, concessions, FAI approval
Programme quality lead	TI One Voice, Brussels (named on the purchase order)	IQC waivers, NCR disposition at MRB, first-delivery audit
Electrical safety reviewer	External, engaged by the programme; **not yet appointed**	Release of Phase 2; any change to a risk-control component
Manufacturer QA manager	Manufacturer	IQC records, FAI report, lot release, CAPA closure
Manufacturer test operator	Manufacturer, named and trained	TST-EEG-004 records per serial

The programme has no full-time quality function. The quality lead is a named programme member with other duties. This plan is written so that it can be operated by one person on the programme side and one QA manager on the manufacturer side.

2. Incoming quality control (IQC)

No lot enters SMT, assembly or kitting until its IQC record is closed. Rejected material goes to a physically separate, labelled quarantine location and is dispositioned under section 7.

2.1 Bare boards

#	Check	Method	Accept	Reject
IQC-B1	Fabrication CoC: IPC-6012 class 2, IPC-A-600 class 2, laminate FR-4 Tg >= 150 C, four layers	Document	Present, lot-specific	Missing or generic
IQC-B2	100 % electrical test certificate to the supplied IPC-D-356A netlist, 156 nets	Document	Present, states “100 %”	Sampled, or netlist not the released one
IQC-B3	Outline 150.0 x 130.0 mm (section 1.1)	Callipers, 2 boards per lot	+/- 0.20 mm	Outside
IQC-B4	Thickness 1.60 mm +/- 10 % over the four-layer stack of section 1.1	Micrometer, 4 points, 2 boards	1.44-1.76 mm	Outside
IQC-B5	ENIG thickness Au 0.05-0.10 um over Ni 3.0-6.0 um	XRF, fabricator report + 1 board per lot	Within band	Outside, or no report
IQC-B6	NPTH holes carry no copper and no mask (ECO-EEG-012): 4 x 3.2 mm at MH1-MH4 and 6 x 1.50 mm DIN retention posts	Visual x10	Bare laminate in every one	Any plating or mask – **hard reject, whole lot**
IQC-B7	Isolation keep-out per DSN-EEG-003 section 3.3, free of copper on **all four layers**	Visual against the fab drawing, plus the inner-layer artwork check of IQC-B13	No copper	Any copper – **hard reject** (S-03)
IQC-B8	J6/J7 hole-row spacing 22.86 mm (ECO-EEG-008)	Pin gauge or CMM, 2 boards	+/- 0.10 mm	Outside – the DevKit will not fit

#	Check	Method	Accept	Reject
IQC-B9	Bow and twist	Surface plate, 2 boards	<= 0.75 %	Outside
IQC-B10	Legend legibility, date code and UL mark on the bottom legend	Visual	Legible, inside the outline, clear of pads	Illegible or over a pad
IQC-B11	**Inner-layer registration.** Microsection of the lot coupon: four copper layers in the order L1 signal / L2 plane / L3 plane / L4 signal; dielectric thicknesses within +/- 10 % of section 1.1; internal annular ring measured at a through via	Microsection, 1 coupon per fabrication lot, report retained	Four layers in the right order; internal annular ring >= 0.025 mm (IPC-6012 class 2)	Wrong layer order, a missing plane, or any internal annular ring below 0.025 mm – **hard reject, whole lot**
IQC-B12	**Layer-to-layer misregistration.** A 0.60 mm pad on a 0.30 mm hole gives a 0.15 mm nominal internal annular ring, so class 2's 0.025 mm minimum survives at most 0.125 mm of misregistration. That is the limit	Fabricator registration report plus the IQC-B11 coupon	<= 0.125 mm at every measured location	Above 0.125 mm

#	Check	Method	Accept	Reject
IQC-B13	**Plane continuity and the plane split.** Inner-layer artwork compared with the released Gerbers for voids, starved etch and slots under the analogue traces; 4-wire resistance corner to corner on each plane region through stitching vias; AGND_REF to DGND open on a bare board, where R90 is not yet fitted	Artwork comparison on every lot; 4-wire DMM on 2 boards, and the insulation tester at 250 V DC for the plane separation	Plane regions <= 0.10 Ohm corner to corner; AGND_REF to DGND **>= 100 MOhm at 250 V DC** (TST-EEG-004 Rev C T0, which owns the limit); no void that breaks the reference under an analogue trace	Any short between the plane regions, or a void under an analogue trace – **hard reject**
IQC-B14	Three fiducials, 1.0 mm copper in a 3.0 mm mask opening, at the positions of section 1.1 (ECO-EEG-020)	Visual, 2 boards	All three present, clean, unmasked	Missing or masked – the placement machine falls back to a vision teach the programme no longer pays for
IQC-B15	**The DRC report that accompanies the fabrication data release has zero open items.** The report issued with the Gerber, drill and IPC-D-356A set names the same design revision as the boards in the lot, and it lists no violation of any class: no unclosed connection, no clearance, width, annular-ring, hole-size, edge, non-plated-hole, isolation-keep-out and no via-keep-out entry, every net	Document. The report is read line by line against the released data set and retained in the IQC pack; the revision on the report is compared with the revision marked on the boards	`VIOLATIONS: 0`, no unclosed connection, and the report's design revision identical to the boards'	Any non-zero violation count, any net that is not one copper island, or a report from a different design revision – **hard reject, whole lot**, and the boards do not enter the build. The Rev B report that accompanies this release, `kicad/EEG-CAR-01_RevB_DRC_report.txt`, is dated **2 September 2026** on its own `Generated` line and records

#	Check	Method	Accept	Reject
	one connected copper island, and both inner planes continuous under the analogue zone			`VIOLATIONS: 0`, all 145 nets connected as one copper island each, and both inner planes continuous, so it satisfies every criterion in this row. *Was: “dated 1 September 2026”* – that was the date of the routing run this plan was first written against, and the report it names was re-generated on 2 September; the date on the report itself is the one an inspector compares, so it is the one stated here. That is not by itself a fabrication release: the data is released for review under RFQ-EEG-002A, the human layout review has not happened, and until it has, no lot can be presented against this row (section 1.1)

Sampling for the dimensional rows is two boards per fabrication lot for Phase 1 and Phase 2, and the c=0 plans of section 6 for Phase 3. Rows IQC-B1, B2, B6, B7, B11, B12, B13 and B15 are checked on **every** board, or on the lot coupon or the released document that represents it – B6, B7 and B13 are safety characteristics, B11 and B12 cannot be seen from the outside of a four-layer board, and B2 and B15 are documents.

IQC-B15 is checked first, because a fabrication release whose DRC report still carries open items is not a release, and no lot built against one is accepted whatever the boards themselves measure.

2.2 Purchased modules

Twelve module types, thirteen assemblies per unit. 100 % inspection for Phases 1 and 2. From Phase 3, the rows marked “sample” drop to the c=0 plans of section 6; the rows marked “every” stay at 100 % for the life of the fleet.

Module	Rate	Check	Accept / reject
ADS1299 breakout x2 (J1/J2/J23, J3/J4/J29)	every	Photograph of the TI package marking, date code and lot recorded; continuity of the 1x12 digital header to the J1 pin order and the 1x10 signal header to the J2 pin order; DAISY_IN and CLKOUT physically exposed; record whether the module needs the J5 jumper; bench power-up and ADS1299 ID register read; AVDD/AVSS measured on-module	Reject on any pinout mismatch, absent DAISY_IN or CLKOUT, ID register mismatch, or AVDD/AVSS outside +2.50/-2.50 V +/- 5 %
ESP32-S3-DevKitC-1-N16R8	every	`esptool flash_id` read back; the module’s own UART USB-C port exercised, because that port and not J26 is the end-of-line flashing route	Accept only 16 MB flash **and** 8 MB PSRAM. The -N16R8 is not substitutable (E-18); mislabelled boards are common
ADuM4160 isolator module	every	Part marking; **host connector type recorded**; barrier visually intact with no component, track or label bridging it; supplier isolation certificate >= 2.5 kV RMS on file, checked once per module lot	Reject on any bridging or a missing certificate. **This is the single safety-critical purchased part** (E-24, S-03). The named candidate presents **USB-B and E-24 asks for USB-C, so E-24 is not met**; the interim answer is the WH-09 USB-B to USB-C panel pigtail, and a USB-B module is accepted only while WH-09 is fitted

Module	Rate	Check	Accept / reject
ATECC608B breakout	every	Factory serial read over I2C and recorded; config zone confirmed UNLOCKED on arrival	Reject a locked part
ES8388 codec	sample	Header pin order against J8 and J9; I2C presence check	Reject the lot on any pin-order mismatch. **AVL-EEG-017 carries this line as OPEN WITH CRITERIA**: no vendor is qualified and an approved sample must be on file before the fleet order.
bq24074-class charger	sample	Header pin order against J12; charge-enable pin responds to CHG_CE	Reject the lot on any pin-order mismatch. **E-23 is met in part, and the two halves must always be stated together.** The thermal-regulation half is met on the charger IC, which is what this incoming check confirms. **The 45 C charge inhibit is not met and cannot be tested**: it needs a cell temperature, and there is no NTC net in design.py and no thermistor way on J12 or J13, so nothing on the carrier can read the cell. **S-04's thermistor-monitored charging is not met either** – the two hardware holes are the same hole (section 14.2; RFQ-EEG-001 Rev E E-23, S-04 and section 12 item 3; AVL-EEG-017 K16)

Module	Rate	Check	Accept / reject
MAX17048 gauge	sample	Header pin order against J12; I2C presence check	As above. Where a combined charger-plus-gauge assembly is supplied, which is the baseline, the two rows are inspected as one against both sets of criteria; where two separate breakouts are supplied, the gauge mounts on MP-01 and its taps are made at the MP-01 end of the J12 Y jumper drawn in ICD-EEG-006 section 3.3
TPS63020-class buck-boost (J25)	sample	Header pin order against J25; output measured at 5.00 V into 0.5 A	Reject the lot on any pin-order mismatch or an output outside 4.90-5.10 V. This is the part ECO-EEG-002 added to make the board power up at all, and it is easy to leave off a kitting list. **AVL-EEG-017 carries this line as OPEN WITH CRITERIA**: no vendor is qualified and an approved sample must be on file before the fleet order.
microSD breakout	sample	Header pin order against J20; one-bit SDMMC lines present	Reject the lot on any pin-order mismatch
Boom preamplifier module (J21, mounted on MP-01)	sample	Header pin order against J21; gain fixed, **automatic gain control absent**; noise floor recorded	**The part is not settled and must not be recorded as settled.** The MAX9814 named in package v1 is an AGC part and E-14 forbids AGC, so it is not approved; the module is specified by interface in ICD-EEG-006 and a fixed-gain part of the MAX4466 class is the preferred route. Reject any module with AGC that cannot be disabled without modifying it

Module	Rate	Check	Accept / reject
Room-microphone module (J28)	sample	Header pin order against J28; hardware mute line exercised	**No module is yet known to meet the hardware-mute requirement of E-15**, so this row is written against an unsourced part; reject anything whose mute is firmware-only
74HC595 breakout (J19)	sample	Header pin order against J19; Q0-Q7 exposed	Reject the lot on any pin-order mismatch. **AVL-EEG-017 carries this line as OPEN WITH CRITERIA**: no vendor is qualified and an approved sample must be on file before the fleet order.
Protected 18650 cell	every	OCV 3.4-3.9 V; internal resistance recorded; protection PCM present and mechanically sound; cell lot recorded; UN 38.3 report and safety data sheet collected per lot (S-04)	Reject a cell outside the OCV band, any cell with a damaged wrap or PCM, any lot without a UN 38.3 report
32 GB industrial microSD	sample	Capacity and part number; full-card write-read verify on the sample	Reject on capacity mismatch or any read error

The boom preamplifier sits on the MP-01 module plate and connects at J21. The boom itself carries the bare electret capsule and its screen on the J18 pigtail. Package v1 put the preamplifier on the boom; `design.py` governs and it does not.

2.3 Carrier passives – the ones that carry the specification

Item	Rate	Check	Accept / reject
R1-R16, **68 k** 0.1 % 25 ppm, Vishay **TNPW060368K0BEEA**	reel label 100 %, plus 5 pieces per reel measured	4-wire DMM	Reject the reel if any piece is outside 0.1 %. These resistors set both the T7 source impedance and the T10 lead-off offset. **A 47 k part in this position is a hard reject**, not a substitution: at 47 kOhm the S-02 single-fault DC current is 53.2 uA against a 50 uA limit and the unit is a safety non-conformance. *Corrected 2026-09-02: this row specified 47 k / TNPW060347K0BEEA and recorded S-02 as not met. ECO-EEG-024 is applied in `tools/design.py`, S-02 is met in the design at 36.8 uA, and E-10 moves to its +/- 1.0 dB branch (TST-EEG-004 T22). The figure is calculated and unsigned-off; see section 14.2 items 1 and 3*
C1-C16, 10 n **C0G** 50 V, Murata **GCM1885C1H103JA16D**	reel label 100 %	Manufacturer part number on the reel	**X7R is a hard reject.** The v1 part number GCM188R71H103KA37D carries the R7 dielectric code and is an X7R part; it is superseded by GCM1885C1H103JA16D (ECO-EEG-019). X7R is +/- 15 % with a voltage and temperature coefficient on top, and the common-mode rejection of the input network is set by the match of the sixteen RC time constants

Item	Rate	Check	Accept / reject
C21, C41, C61, 100 n **25 V** and C22, C42, C62, 220 n 25 V Sallen-Key capacitors, Murata GCM188R71E104KA57D and GCM188R71E224KA64D	reel label 100 %	Manufacturer part number on the reel	**Both are X7R, declared, with a stated 15 % capacitance tolerance over temperature.** A 100 nF C0G in 0603/50 V is not a stocked part, so these are X7R by decision and not by accident (ECO-EEG-019). The consequence is stated in section 4.3 row FA-09: the envelope corner cannot be held inside E-11's +/- 10 % by the fitted dielectric, and **E-11's 50 Hz +/- 10 % half is therefore not met** – TST-EEG-004 T12e measures f0 per unit against 42 to 58 Hz, which is the band TST-EEG-004 sets against the parts approved here and not a widening of any E-11 limit that was ever written (TST-EEG-004 section 16 item 16). IQC cannot inspect its way round this: a reel that conforms to its own part number still gives a corner outside E-11. *Both are 25 V parts: the "50 V" of the previous issue of this row was wrong against `design.py`, which governs, and against the 71E rating code in both part numbers*
Envelope-chain thin films (R20-R28, R40-R48, R60-R68, all 0.1 %), U1-U3 **OPA4376AID (SOIC-14), TI OPA4376AIDR**, U7 TLV3201AIDBVR	reel and tube labels	Part number and date code	Reject on any substitution not carrying an approved ECO. The v1 "OPA4376AIPW" TSSOP part number is superseded; `design.py` fits the SOIC-14

Item	Rate	Check	Accept / reject
D1-D16, **BAV99**, Nexperia BAV99,215	reel label 100 %	Manufacturer part number on the reel	**BAT54S is a hard reject in the D1-D16 positions.** Schottky leakage across a 68 kOhm series resistor is an offset error on a 10 uV input. BAT54S is fitted only at D20, D40 and D60, the envelope rectifiers
R94, R95, 4k7 1 % I2C pull-ups	reel label 100 %	Part number	Added by ECO-EEG-021 so that the bus does not depend on whatever the modules happen to carry

Verification and sign-off. IQC is performed by the manufacturer's goods-in inspector and signed by the manufacturer QA manager. The programme quality lead countersigns the Phase 1 IQC pack before the first build and audits it thereafter under section 13.

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3. In-process inspection points

IP	Point in the flow	What is checked	Rate	Record
IP-1	Solder paste, before placement	Stencil part number and revision; paste alloy and lot; SPI or a 3-board first-off visual; the three fiducials read by the placement machine	First-off + 1 per shift	Board serial, paste lot
IP-2	After reflow, before THT	AOI to IPC-A-610 class 2; polarity of D1-D16 (BAV99), D23, D24; **BAT54S pin 3 to the op-amp output** on D20, D40 and D60 (ECO-EEG-005); U1-U3 and U7 pin 1	100 %	AOI report per board

IP	Point in the flow	What is checked	Rate	Record
IP-3	THT and socket insertion	**33 through-hole parts, 236 plated holes**, plus the 10 non-plated holes of IQC-B6 which stay bare; socket strips seated flush, standoff <= 0.5 mm; pin 1 square pad against the legend; J15-J17 DIN sockets perpendicular with retention posts engaged	100 %	Operator initials
IP-4	Star points and links	The star-point rule of DSN-EEG-003 section 3.3: R90 fitted exactly once; R91 fitted exactly once; **no wire link or solder bridge across either**; R92 and R93 fitted by default; R89 not populated	100 %	Tick sheet
IP-5	Isolation keep-out	The strip of DSN-EEG-003 section 3.3 free of solder, flux residue, adhesive and any part body, on both outer layers	100 %	Photograph retained
IP-6	Cleanliness	J-STD-001 class 2; no-clean residue acceptable; if an aqueous process is used, ionic cleanliness per the manufacturer’s own qualified limit	1 per lot	Certificate

IP	Point in the flow	What is checked	Rate	Record
IP-7	Module fitting	Modules on the MP-01 plate (146.0 x 126.0 x 3.0 mm), keyed ribbon jumpers per ICD-EEG-006 with the WH-KEY-01 printed keying shrouds fitted; four M3 x 18 mm nylon female-female standoffs and eight M3 x 6 nylon pan screws; the DevKit inserted directly into J6/J7 with no bent pins	100 %	TST-EEG-004 T2
IP-8	Harness build	Continuity of every conductor on the 12-way screened electrode cable to J14 and the 10-way light cable to J30; screen continuity; screen landed at one end only; insulation resistance	100 %	Wire-list tick sheet per WH-EEG-008
IP-9	Printed parts, goods-in from the print bureau	Dimensional check of the HM-04 bayonet, collar and hatch fits against the FIT-01 coupon (60.0 x 24.0 x 10.0 mm, bores 9.20 / 9.35 / 9.15 mm); powder fully removed from internal channels	First-off per build, then sampled	Coupon result
IP-10	Pod closure	Stack height measured against the 55.5 mm internal depth of POD-P1 (49.1 mm calculated, margin 6.4 mm); openings gasketed or recessed; **no LED opening – M-02 is withdrawn and the pod carries no indicator**; no gel path to the board; label applied per PKG-EEG-015	100 %	T18

IP	Point in the flow	What is checked	Rate	Record
IP-11	ESD	Operator wrist strap and mat verified at shift start; EPA verified per ANSI/ESD S20.20	Per shift	ESD log

IP-11 exists because the ADS1299 front end, the three OPA4376 quads at U1-U3 and the sixteen high-impedance input networks are exactly the parts that electrostatic damage degrades silently into a noisier channel that still passes a functional check.

The 18 mm standoff at IP-7 is also a safety characteristic. The slant path from carrier copper, over the edge of the isolation keep-out and up the standoff to any host-side conductor on MP-01, is at least 18 mm, which is more than twice the 8 mm the safety case asks for. This is what closes RISK-EEG-011 SR-08, so a shorter standoff is not a substitution the line may make.

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4. First-article inspection (FAI)

An FAI is required before the first batch of each phase, and again after any ECO that touches the board, a printed part, the harness or the firmware pin map. It follows the AS9102 three-form structure, adapted: this is not an aerospace part and no AS9102 compliance is claimed. FAI is performed on the **first assembled carrier** and the **first complete kit**, and is approved in writing by the programme technical lead before the balance of the batch is built.

4.1 Form 1 – part number accountability

Field	Entry
Part number / revision	EEG-CAR-01 Rev B (carrier); TIOV-EEG-KIT Rev A (kit)
Part name	Sixteen-channel research EEG carrier / EEG field kit
Serial number	First unit of the batch, programme serial as allocated (section 9)
FAI type	Full (first article) or partial (state which characteristics are re-verified and why)
Drawing / data set	design.py Rev B, released four-layer Gerber and drill set, IPC-D-356A netlist, CPL, ASM-EEG-007, WH-EEG-008, kit BOM, and the DRC report issued with them. No FAI is opened against a data set whose DRC report carries open items (IQC-B15)
Detail parts and assemblies	211 reference designators, of which 186 are purchased placements; 12 purchased module types in 13 assemblies; the printed parts listed in PARTS-EEG-019 section 2; harness; case and foam

Field	Entry
Reason for FAI	New design / ECO number / process change / vendor change / lapse > 24 months

4.2 Form 2 – product accountability: materials, processes, testing

Item	Requirement	Evidence
Laminate and stack-up	FR-4, Tg >= 150 C, four layers, 1.60 mm +/- 10 %, 1 oz outer and 0.5 oz inner copper, per section 1.1	Fabricator CoC, lot number, stack-up drawing as built
Inner layers	Microsection coupon per IQC-B11; misregistration per IQC-B12; plane continuity per IQC-B13	Microsection report, registration report, 4-wire results
Surface finish	ENIG, Au 0.05-0.10 um over Ni 3.0-6.0 um	XRF report
Solder alloy	Stated by the manufacturer (SAC305 assumed unless declared otherwise); flux type	Alloy CoC, reflow profile record
Reflow profile	Board-specific profile with the thermocouple trace	Profile record
Through-hole process	Hand or selective solder, J-STD-001 class 2, 236 plated holes	Process sheet
Printed parts	MJF PA12, dyed; orientation, powder removal, finishing per the print specification	Print bureau CoC, FIT-01 coupon results
Skin-contact materials	ISO 10993-5 / -10 supplier declarations for cups, ear clips, TPU pads, the HM-06 chin strap and the HM-03 occipital yoke (S-05). The A-03 headband is withdrawn as a kit item, so there is no declaration to collect against it	Declarations on file
Cell	Protected 18650, UN 38.3 report per lot (S-04)	Report reference recorded in the DHR
Special processes with no qualification	Laser marking. **Conformal coating is not applied**: the decision is taken, not deferred (section 14.2)	–
Functional testing	TST-EEG-004 Rev C, all steps, on the FAI unit	Full record, all numeric values

Item	Requirement	Evidence
Type tests, once per model rather than per unit	Maximum acoustic output <= 100 dB SPL on an artificial ear at any commanded level (E-29), with the firmware codec volume clamp set to the value measured at calibration; the ADuM4160 supplier's 2.5 kV RMS isolation certificate	Type-test report; supplier certificate

E-29 is a new requirement and it exists because the calculated full-scale output of the headphone chain is about 110 dB SPL, which is above the limit. Until the type test is run on the shipped headphone model, the clamp value is provisional.

4.3 Form 3 – characteristic accountability

Every characteristic below is measured on the first article and recorded with its actual value, not a tick. "Calculated" marks a design value that has never been measured on hardware. The requirement values are not restated here: they are in DSN-EEG-003 section 3.2 for the board and in section 1.1 above for the two layout changes. What this form adds is the gauge, the sample and the signature.

Bare board

#	Characteristic	Requirement	Method / gauge	Signed by
FB-01	Outline length	150.0 +/- 0.20 mm	Callipers	QA
FB-02	Outline width	130.0 +/- 0.20 mm	Callipers	QA
FB-03	Thickness	1.60 mm +/- 10 %	Micrometer, 4 points	QA
FB-04	Layer count and order	**4: L1 signal, L2 reference plane, L3 reference plane, L4 signal**	Micro-section of the lot coupon	QA + programme
FB-05	Outer copper	1 oz / 35 um	Fabricator report	QA
FB-06	Inner copper	0.5 oz / 17 um	Fabricator report, confirmed on the micro-section	QA
FB-07	Dielectric thicknesses	prepreg 0.200 / core 1.065 / prepreg 0.200, each +/- 10 %	Micro-section	QA
FB-08	ENIG Au / Ni	0.05-0.10 um / 3.0-6.0 um	XRF	QA

#	Characteristic	Requirement	Method / gauge	Signed by
FB-09	Minimum track	>= 0.20 mm at the narrowest conductor named in `kicad/EEG-CAR-01_RevB_DRC_report.txt`. The routing released for review measures 0.20 mm at its narrowest, and 36 of its 169 relaxed connections take a conductor narrower than the 0.25 mm preferred width. The report is the authority for which conductor is measured, and it is accepted for a fabrication release only under IQC-B15	Optical comparator	QA
FB-10	Minimum clearance	>= 0.20 mm	Optical comparator	QA
FB-11	Electrode-net clearance	>= 0.35 mm to anything else	Optical comparator on two named nets	QA + programme
FB-12	Via construction	**Through vias only**: 0.60 mm pad / 0.30 mm finished hole, tented both sides, no blind, buried, back-drilled, filled or plugged vias	Pin gauge, visual, micro-section	QA
FB-13	Plated hole sizes	0.30 (vias) / 0.90 / 1.00 / 1.20 / 1.70 mm, +/- 0.08 mm	Pin gauges	QA
FB-14	NPTH 3.2 mm at MH1-MH4	Positions (5,5), (145,5), (5,125), (145,125) in design coordinates; 6 mm copper keep-out on all four layers	Callipers, visual	QA
FB-15	NPTH 1.50 mm DIN retention posts x6	No copper, no mask (ECO-EEG-012)	Visual x10	QA + programme

#	Characteristic	Requirement	Method / gauge	Signed by
FB-16	Isolation keep-out	Per DSN-EEG-003 section 3.3, no copper on any of the four layers	Visual against fab drawing and inner-layer artwork	Programme + safety reviewer
FB-17	Reference planes	AGND_REF left of x = 62 mm and DGND right of it, **on both L2 and L3**, stitched	Inner-layer artwork against the released Gerbers	Programme
FB-18	Fiducials	Three, 1.0 mm copper in a 3.0 mm mask opening, at the section 1.1 positions	Visual, optical comparator	QA
FB-19	J6-J7 row spacing	22.86 mm +/- 0.10 mm (ECO-EEG-008)	Pin gauge or CMM	QA
FB-20	Electrical test	100 % to the IPC-D-356A netlist, 156 nets	IPC-9252 flying probe or bed of nails	QA
FB-21	Internal annular ring	>= 0.025 mm, IPC-6012 class 2	Micro-section (IQC-B11)	QA + programme
FB-22	Layer-to-layer misregistration	<= 0.125 mm, derived in IQC-B12	Registration report + micro-section	QA + programme
FB-23	Plane continuity	<= 0.10 Ohm corner to corner within each plane region, both inner layers	4-wire DMM through stitching vias	Programme
FB-24	Plane separation	AGND_REF plane to DGND plane **>= 100 MOhm on the bare board, R90 not fitted** (TST-EEG-004 Rev C T0, which owns the limit)	Insulation tester at **250 V DC**	Programme
FB-25	Bow and twist	<= 0.75 %	Surface plate	QA
FB-26	Mask and legend	Green LPI both sides, white legend both sides, legible	Visual	QA

#	Characteristic	Requirement	Method / gauge	Signed by
FB-27	Marking	Fabricator date code and UL mark on the bottom legend, inside the outline, clear of all pads	Visual	QA

Assembled carrier

#	Characteristic	Requirement	Method	Signed by
FA-01	Bill of material as built	211 designators; 186 purchased placements of which 33 are through-hole; R89 not populated; R90, R91, R92, R93 per section 3 IP-4; R94 and R95 fitted (ECO-EEG-021)	Ref-by-ref against the released BOM	QA
FA-02	Workmanship	IPC-A-610 class 2, no waivers	Visual + AOI (T1)	QA
FA-03	R1-R16 value	**68 k** 0.1 % as fitted (ECO-EEG-024, applied in `tools/design.py` 2026-09-02), measured on 4 of 16 in circuit or on the reel sample. At this value S-02's single-fault DC current is **36.8 uA calculated against 50 uA and the limit is met**; the measured resistor value is recorded, and the E-10 band applied at TST-EEG-004 T22 follows from it (+/- 1.0 dB at 68 k, +/- 0.5 dB at 47 k). *Corrected 2026-09-02: was “47 k as fitted on the Phase 1 prototypes … S-02 is not met at this value”*	4-wire DMM	QA + programme
FA-04	C1-C16 dielectric	C0G, part number GCM1885C1H103JA16D	Reel label, retained photograph	QA

#	Characteristic	Requirement	Method	Signed by
FA-05	D20/D40/D60 orientation	BAT54S pin 3 at the op-amp output (ECO-EEG-005); D1-D16 are BAV99 and are checked at IP-2	Visual x10	QA
FA-06	V5V rail	5.00 V, measured at TP16	DMM	Test operator
FA-07	AVDD / AVSS	+2.50 V / -2.50 V at TP10 / TP11	DMM	Test operator
FA-08	AGND_REF	Analogue mid-rail present at TP13, not tied to DGND anywhere but R90	DMM + continuity	Programme

#	Characteristic	Requirement	Method	Signed by
FA-09	Envelope corner	f0 = 48.77 Hz *calculated*, Q = 0.7416 *calculated*, from R = 22 k, C_gnd = 100 nF, C_fb = 220 nF; AC-coupling corner 1.59 Hz *calculated* from 10 uF into 10 k (ECO-EEG-027), against E-11 restated as <= 2 Hz. Measured f0 recorded per unit at TP7, TP8, TP9 against **TST-EEG-004 T12e's 42 to 58 Hz**. **The 50 Hz +/- 10 % half of E-11 is not met by the fitted parts and this row does not grade it as met**: C21/C41/C61 and C22/C42/C62 are both X7R at 15 %, f0 goes as 1/sqrt(C1 C2), so a common 15 % moves it to 42.4 to 57.4 Hz – a spread wider than the +/- 10 % E-11 asks for wherever the centre sits. T12e's 42 to 58 Hz is the band this package sets against the parts it approved; it is not a widened E-11 limit, because no f0 limit existed to widen (TST-EEG-004 section 16 item 16, RFQ-EEG-001 Rev E section 12 item 15). The measured value is what the record carries	Swept sine	Test operator + programme

#	Characteristic	Requirement	Method	Signed by
FA-10	Comparator threshold	52 mV *calculated*, hysteresis ~5 mV *calculated*; measured value recorded (E-12). Record which comparator configuration was built: `design.py` Rev B powers U7 from AVDD/AVSS, and ECO-EEG-023 re-powers it from DVDD3V3/DGND with the inputs re-referenced to a DVDD3V3/2 divider and the envelope AC-coupled into it. **ECO-EEG-023 is not in the released design.py and must be checked by the safety and layout reviewer before it is cut in**	Ramp at the ENV_STIM node	Test operator + programme
FA-11	Input network loss at 100 Hz	**0.75 dB** *calculated* at the fitted **68 k** / 10 n; measured value recorded against E-10's **+/- 1.0 dB** branch. At the superseded 47 k the loss was 0.36 dB against +/- 0.5 dB. *Corrected 2026-09-02: ECO-EEG-024 is applied and the two figures have swapped roles.* The arithmetic is in RISK-EEG-011 section 4 and is not repeated here	Swept sine through the T7 fixture	Programme
FA-12	Noise floor	**0.31 uV RMS** *calculated* at the fitted 68 k (0.27 uV at the superseded 47 k), both against the 1.0 uV limit of E-03; measured per T8. *Corrected 2026-09-02 to name the fitted value first.*	TST-EEG-004 T8	Test operator

#	Characteristic	Requirement	Method	Signed by
FA-13	Test access	TP1-TP18 present and probeable, and the 1x6 UART debug header at J26 fitted, way 6 being the spare NC_GPIO0 because GPIO0 is committed to LED_SR_LATCH (E-28, ECO-EEG-009). No JTAG connector is fitted and none is required	Visual + probe	QA
FA-14	Full TST-EEG-004 record	Every step, every numeric value	TST-EEG-004 Rev C	Test operator + QA

Harness and kit

#	Characteristic	Requirement	Method	Signed by
FH-01	Electrode cable	12 conductors, screened, to J14; from-to per WH-EEG-008	Continuity, conductor by conductor	QA
FH-02	Light cable	10 conductors to J30	Continuity	QA
FH-03	Screen	Continuous, landed at the pod end only (R91)	Continuity + insulation resistance	QA
FH-04	Insulation resistance	Conductor to conductor and conductor to screen, at the voltage stated in WH-EEG-008	Insulation tester	QA
FH-05	Barrier insulation, per unit	**500 V DC insulation resistance across the isolation barrier.** There is no per-unit hipot: the 2.5 kV RMS type test is the module supplier's certificate, checked once at incoming inspection	Insulation tester	QA + programme

#	Characteristic	Requirement	Method	Signed by
FK-01	Helmet fit	Adjustment range and electrode positions per DSN-EEG-002 Rev E	Headform gauge	QA
FK-02	Pod	Closed, gasketed, no gel path, no indicator opening (M-02 withdrawn)	Visual	QA
FK-03	Label	Serial, hardware revision, key fingerprint, “research instrument – not a medical device”, programme URL (M-03)	Read back and compare with the calibration record and the USB iSerial from T5	QA + programme
FK-04	Case and foam	Cut-outs match every item; labels legible (M-05, M-06); the printed calibration certificate in the case lid pocket beside the quick-start card	Pack and unpack once	QA
FK-05	Packing list	Every line ticked (T18)	Tick sheet	QA
FK-06	Consumables	Quantities per A-05; expiry dates recorded. A-03 is the chin strap HM-06 and the occipital yoke HM-03, not a headband	Count	QA
FK-07	Cell	Installed cell lot and OCV recorded; **no spare cell in the case**, per PKG-EEG-015 section 7	Record	QA + programme

5. Sampling: what it is for, and what it is not for

At 2, 10, 25 and 50 units, **sampling the assembled units makes no sense and is not practised**. Every unit gets every step of TST-EEG-004, plus the label and packing checks. The arithmetic is plain: under ISO 2859-1 / ANSI-ASQ Z1.4 general inspection level II, a lot of 10 at AQL 1.0 sends you to a sample size larger than the lot itself, and a sample of 13 from a lot of 10 costs more to administer than testing all 10. Two further reasons make 100 % test the right choice here independently of cost: the per-unit numbers from T7, T8, T9, T10, T12, T13 and T17 are *study metadata* that must travel with each instrument, so they have to be produced for every unit anyway; and a fleet of 25 to 50 units has no statistical population to speak of.

Sampling is therefore used for exactly three things:

1. **Consumables and multi-piece accessories** bought by the hundred or thousand, where 100 % inspection is genuinely wasteful (section 6).
2. **Reel-level and lot-level characteristics** of passives, where the reel is the unit of quality, not the piece. The four-layer coupon characteristics of IQC-B11 and IQC-B12 belong to the same class: the fabrication lot is the unit of quality, because registration is a property of the press cycle and not of the individual board.
3. **Destructive or slow measurements** – electrode DC offset and drift, cell internal resistance, print coupon dimensions, the inner-layer micro-section – where measuring every piece would consume the piece or the schedule.

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6. AQL plans for the fleet builds

The named standard is **ISO 2859-1 / ANSI-ASQ Z1.4, general inspection level II**, so that a manufacturer can operate it with the tables they already have. Where a plan is stated below, the programme uses **zero-acceptance (c = 0)** plans and derives the sample size from the binomial directly, so the plan can be checked without a table:

> $n = \ln(0.10) / \ln(1 - RQL)$, rounded up. Accept on 0 defects, reject on 1. RQL is the fraction defective at which the lot is accepted with only a 10 % probability.

Class	Defect examples	RQL	n (calculated)
Critical	Anything affecting S-01 to S-08: a bridged isolation barrier, copper inside the keep-out on any of the four layers, an unprotected electrode lead, a cell without a UN 38.3 report, a missing protection resistor	–	**100 %, accept on 0.** Sampling is not used for critical characteristics
Major	Function-affecting: wrong dielectric, wrong header pinout, mis-cut foam, missing consumable line, illegible label	5 %	45
Minor	Cosmetic: legend smudge, dye variation on a printed part, scuffed case	10 %	22

Applied to the fleet builds:

Item	Qty at 10 kits	Qty at 25	Qty at 50	Plan
Sintered Ag/AgCl cup electrodes	100	250	500	100 % visual; **n = 45, c = 0** for DC offset and drift
Ag/AgCl ear clips	20	50	100	100 % visual; n = 22 for offset
HM-04 electrode bodies (printed)	100	250	500	n = 45 dimensional against FIT-01 (bayonet, collar); 100 % visual for powder residue
Electrode springs 3-6 N	100	250	500	n = 22 force check
Disposable EMG snap pads	300	750	1500	n = 22, plus 100 % expiry-date check on the packs
Saline wipes, paste, prep gel	10-20 packs	25-50	50-100	100 % expiry-date check; n = 22 on pack integrity
Headphones	10	25	50	**100 %** functional – the calibrated output level is only transferable if the model is identical (A-04, restated as 32 to 64 Ohm)
USB cables, EU chargers	10-20	25-50	50-100	n = 22 functional; 100 % CE marking check
Travel cases and foam inserts	10	25	50	100 % against the CASE-00 cut file; the foam is the item most likely to be wrong
0603 passives	reels	reels	reels	Reel label 100 %; 5 pieces measured per reel
Bare-board fabrication lots	1-2 lots	2-3	3-5	Lot coupon 100 % for IQC-B11 to IQC-B13; dimensional rows n = 2 boards per lot

6.1 Switching rules

Normal inspection is the default. Switch to **tightened** (RQL halved, so n = 91 for major and n = 45 for minor) when 2 of 5 consecutive lots are rejected on the same characteristic. Switch back to normal after 5 consecutive lots accepted under tightened inspection. **Reduced inspection is not used.** State plainly: with three or four production lots in the entire programme, the switching rules will rarely fire. They are written down so that a repeated defect cannot be normalised, not because the programme expects to run them.

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7. Non-conforming material

7.1 Failure codes

Codes are keyed to the test that found the fault, so that first-pass yield can be reported per step. The code is **F-T**, where nn is the step number in TST-EEG-004 Rev C. That document owns the step numbers and their names, and they are not restated here; a code is only valid if the step exists in TST-EEG-004 Rev C. IQC failures take the code **F-IQC-**, using the row identifiers of section 2 – for example F-IQC-B12 for a misregistered fabrication lot.

7.2 Disposition

Disposition	When it applies	Limits
Re-seat and re-test	Module seating and keying faults only	Maximum 2 insertion cycles; cycles logged against the socket
Rework and re-test	A single joint or a single passive	Maximum 2 hand-rework cycles per site, IPC-7711/7721. R1-R16 re-measured for 0.1 % after any rework. More than 2 rework sites in the analogue zone (x < 62 mm) sends the board to scrap
Module replacement	Any module fault	**Mandatory full re-run of T7, T8, T9, T10 and re-write of the calibration constants.** A module may never be swapped after provisioning without re-characterisation
Quarantine, no rework	Any isolation-barrier damage near J10; any DIN socket (J15-J17) rework; a dropped or deformed cell; a suspected counterfeit ADS1299; any bare board that failed IQC-B11, B12 or B13, because an inner-layer defect cannot be reworked	Escalated to the programme; feeds RISK-EEG-011

Disposition	When it applies	Limits
Scrap	Board fails the above limits, or fails the bare-board electrical test	Scrap authority: manufacturer QA manager
Use as is / concession	**Prohibited** for T7, T8, T9, T10, T13, for any four-layer characteristic in IQC-B11 to B13, and for any S-requirement failure	Any other concession needs the programme technical lead's written approval, an expiry date and an ECO number

7.3 Material review board

Composition: manufacturer QA manager (chair), manufacturer process engineer, programme quality lead. Any deviation affecting a value recorded in the calibration record is escalated to the programme technical lead. Re-test rules: a unit may be re-tested at most twice on the same step; the second failure raises a non-conformance report. **Every attempt is retained in the record. Results are never overwritten.**

7.4 Records and delivery

Failed units' records are delivered with the batch, carrying the failure code and the disposition. The lot summary reports first-pass yield per test step, so that the programme can see which step drives yield before scaling from 10 units to 50.

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8. Corrective action

A corrective action is raised on: any critical-characteristic escape; any repeat of the same failure code on 3 units in one lot; any lot rejection at IQC; any customer (programme) complaint after delivery; and any field return whose root cause is a build fault.

The route is an 8D-style report, on one page:

1. Team and contact.
2. Problem statement with the failure code, serials affected and the number found.
3. Containment: what is quarantined now, and how far back the suspect population reaches (state the serial range).
4. Root cause, with the evidence. "Operator error" is not accepted as a root cause without a process reason behind it.
5. Chosen corrective action.
6. Verification that the action works, with data.
7. Preventive action: the process change, the ECO number if a document or the design changes.
8. Closure, signed by the manufacturer QA manager and the programme quality lead.

Target: containment within 2 working days of raising, root cause within 10, closure within 30. Any corrective action touching a risk-control component or the isolation architecture re-triggers the electrical safety review of RFQ section 9.2 before further units ship.

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9. Device history record (DHR)

One record per programme serial. The label artwork is carried in PKG-EEG-015 section 5 and the change register in ECO-EEG-016; the definitions below are the ones this plan records against, so that the label, the record and the firmware agree.

- **Programme serial:** `TIOV-B-nnnn` – programme prefix, hardware revision letter, four digits. Phase 1 uses 0001-0009, Phase 2 0010-0099, Phase 3 0100-0999. Serials are allocated by the programme in blocks and issued with the purchase order. The same string appears in the label text, the Data Matrix, the USB `iSerialNumber`, the calibration record and the packing list; if any two disagree the unit is quarantined.
- **ATECC608B device serial:** the 9-byte factory serial, read at boot, recorded in the device history record and printed on the label. It is **not** written into the USB `iSerialNumber`; that field carries the programme serial `TIOV-B-nnnn` alone (F-04, RUL-EEG-021 section B). The two are checked against each other at T5b.
- **Key fingerprint:** as defined in FW-EEG-001 section 7. The definition is not repeated here. This is what T6 records and what T16 and the label are checked against.

9.1 What is filed per serial, at build

Group	Content
Configuration	Purchase order; BOM revision built to; design.py / Gerber revision and the DRC report reference; ECO numbers applied; firmware image version and SHA-256 hash
Material	Bare board fabrication lot and date code, with the micro-section and registration report references for that lot; ADS1299 module #1 and #2 vendor lot **and** TI device date code and lot; ADuM4160 module lot and isolation certificate reference; ATECC608B factory serial; cell lot, OCV at goods-in and UN 38.3 report reference; R1-R16 reel lot; C1-C16 reel lot
Process	Reflow profile record; ESD log for the shift; operator identities; station identity
Test	The complete TST-EEG-004 record, every numeric value, including failed attempts; test-equipment asset numbers and calibration certificate references
Provisioning	Provisioning log; exported public key; fingerprint; VID/PID and hardware revision written. **eFuses are not burned on Phase 1 prototypes**: secure boot and flash encryption are enabled from Phase 2, so the Phase 1 units run unsigned images and T25 is a Phase 2 step, not a Phase 1 gate
Release	FAI report reference (first unit of the batch); deviations and concessions applied; final assembly and packing check; dispatch record

9.2 What is added per turnaround

Refurbishment record per SVC-EEG-013; parts replaced with their lots; cell cycle count and OCV; disinfection record; re-test results if any; the participant assignment period (held by the programme only, and kept separate from the manufacturer's copy for data-protection reasons).

9.3 Where it lives and for how long

Copy	Held by	Format	Retention
Master	TI One Voice programme, Brussels	Machine-readable calibration record (one file per unit) plus the scanned paper pack	Life of the fleet **plus 10 years**
Manufacturer copy	Manufacturer	Their own QMS format	7 years from dispatch, minimum
Delivery copy	Travels with the unit, in the case lid pocket	Printed calibration record	Life of the unit; reissued at each refurbishment

Acceptance in Brussels (RFQ section 9.3) is conditional on receipt of the per-unit record. A batch delivered without records is not accepted.

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10. Traceability

RFQ-EEG-001 section 10 requires "date code and lot for the two converters and the cell recorded per unit". This plan states how.

Item	What is recorded	How it is captured	Where it goes
ADS1299 module #1 (J1/J2/J23)	Module vendor, module board revision, module lot; TI ADS1299 package marking, date code and lot; a photograph of the marking	At IQC, before the module is fitted	DHR, material group
ADS1299 module #2 (J3/J4/J29)	As above	As above	DHR
Protected 18650 cell	Cell manufacturer, cell lot, OCV at goods-in, UN 38.3 report reference	At IQC	DHR + PKG-EEG-015 shipping file
ADuM4160 module	Vendor, module lot, isolation certificate reference, host connector type and whether WH-09 is fitted	At IQC	DHR

Item	What is recorded	How it is captured	Where it goes
ATECC608B	Factory 9-byte serial	Read over I2C at IQC and again at provisioning; both recorded	DHR + label
Bare board	Fabrication lot and fabricator date code from the bottom legend; micro-section, registration and plane-continuity report references for the lot	At board goods-in	DHR
R1-R16, C1-C16	Reel lot codes	At kitting for SMT	DHR
Firmware	Image version and SHA-256	At flashing, over the DevKit's own UART USB-C port	DHR + calibration record

The record must be able to answer, without opening a box: which units carry a given ADS1299 lot; which units carry cell lot X; which units came from a given bare-board fabrication lot; which units are on firmware version Y; which participant had which unit and when. Those five queries are the acceptance test for the DHR format. The fabrication-lot query is new in Rev B and exists because an inner-layer defect is invisible from outside the board, so the only way to bound a suspect population is by press lot.

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11. Supplier quality and change notification

11.1 Supplier classes

Class	Items	Requirements
A – safety or characterisation critical	ADuM4160 module (J10); ADS1299 modules (J1-J4); protected 18650 cell; sintered Ag/AgCl cups and ear clips; DIN 42802 sockets J15-J17; the EU charger; the bare-board fabricator	Named vendor and part number; controlled revision; schematic or dimensioned drawing; certificate of conformity per lot; **change notification 90 days before any board revision, process change or site move**; isolation rating certificate for J10; ISO 10993 declarations for skin-contact items; for the fabricator, the stack-up and registration process are part of the controlled configuration and may not change without notification
B – function critical	The remaining module types; print bureau; travel case and foam; headphones	CoC per lot; change notification before any revision; part number pinned

Class	Items	Requirements
C – commodity	Passives, socket strips, fasteners, consumables	CoC; traceable lot codes for R1-R16, C1-C16 and D1-D16

11.2 Rules that fall out of this

- The permissive language in the v1 kit BOM – “Chinese equivalent”, “generic”, “certified generic”, “generic ADS1299 breakout with same header pinout” – is superseded. Every line in AVL-EEG-017 Rev C is either a named approved part or an explicit open-with-criteria entry that states the criteria.
- **J15 to J17 have no confirmed part and that is a live procurement risk.** `design.py` names Stäubli SLB1,5-F / LB-I1,5 as a class, not a confirmed PCB part. A touch-proof 1.5 mm socket with a PCB-mount signal pin and two 1.5 mm retention posts must be sourced and first-articled before Phase 2. The criteria are: touch-proof 1.5 mm to DIN 42802, finger-safe per IEC 60601-1, colour-coded, stated current rating and mating force, and a sample submitted to the programme for approval. AVL-EEG-017 carries a 12-week lead-time risk against it.
- The four parts RFQ section 10 declares non-substitutable – the two ADS1299 modules, the ATECC608B breakout, the ADuM4160 module and the ESP32-S3-DevKitC-1 – may not be changed by any route other than a substitution request approved by the programme technical lead and issued as an ECO. ICD-EEG-006 names only two of the four; RFQ-EEG-001 Rev E sits above ICD-EEG-006 in the precedence order and all four apply.
- **Counterfeit avoidance for the ADS1299.** Purchase only through the module vendor or an authorised TI distributor with a documented chain. No broker or open-market purchases. The date code and lot are retained per unit (section 10). A T7 gain or T8 noise outlier is treated as a possible counterfeit indicator and quarantined, not reworked. X-ray is retained for Phase 1; its scope is open (section 14.2).
- **Substitution request form:** the part, the proposed alternate, the reason, a datasheet comparison against every affected E-nn / S-nn requirement, the test evidence, the impact on RISK-EEG-011, the programme’s approval signature, and the resulting ECO number.

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12. Gauge R&R for the gain and noise measurements

T7 (gain within 0.5 % at the 1 mV point) and T8 (noise <= 1.0 uV RMS) produce the study’s primary instrument characterisation. If the measurement system is not capable, unit-to-unit differences in the record are fixture drift and operator technique, which is precisely the confound RFQ section 9 exists to avoid.

12.1 Study design

Three units x three operators x three repeats, run on the first production fixture set FIX-01 to FIX-04 of JIG-EEG-009, before the first fleet batch, and repeated whenever a fixture is rebuilt or a second set is introduced. Reported as %GRR against the tolerance band, where %GRR = 6 sigma divided by the full width of the band. **Requirement: %GRR <= 20 %.**

12.2 What the arithmetic already says about T7

The estimator itself has a floor. For a sinusoid of amplitude A measured by coherent detection over T seconds against a white noise density n, the standard deviation of the amplitude estimate is `sigma = n / sqrt(2T)`. Taking E-03's 1.0 uV RMS over the 0.5-70 Hz band gives n = 0.120 uV/sqrt(Hz), and at T = 60 s, sigma = **0.0110 uV (1 sigma)**. The expanded uncertainty at **k = 2** is 0.0219 uV. JIG-EEG-009 section 1.4's derivation governs; the v1 audit and Rev A of this plan quoted 0.219 % at the 10 uV point as a 1 sigma figure, and that label was wrong – 0.219 % is the k = 2 value and 1 sigma is 0.110 %. All *calculated*:

T7 point	Applied amplitude	1 sigma at 60 s	U at k = 2	6 sigma	Limit	%GRR from the noise term alone
1 mV	1000 uV	0.0011 %	0.0022 %	0.0066 %	+/- 0.5 %	0.7 %
100 uV	100 uV	0.011 %	0.022 %	0.066 %	+/- 0.5 %	6.6 %
10 uV	10 uV	0.110 %	0.219 %	0.657 %	+/- 0.5 %	**66 % – not capable**
10 uV	10 uV	0.110 %	0.219 %	0.657 %	+/- 5 %	6.6 %

So T7's gain figure as written cannot be measured to 0.5 % at the 10 uV point in a 60 s record, however good the fixture is. Lengthening the record to 240 s only halves sigma (0.055 %, 6 sigma = 0.33 %, still 33 % and still above the 20 % requirement). **The plan therefore splits T7:**

- The **released gain constant** per channel is taken at the 1 mV point, limit +/- 0.5 %.
- The 100 uV point is a linearity check, limit +/- 0.5 %.
- The 10 uV point is a linearity check with a limit of **+/- 5 %**, which gives %GRR = 6.6 % from the noise term and leaves room for the fixture.

The +/- 5 % limit is the ruled value for the 10 uV point. TST-EEG-004 T7 and JIG-EEG-009 sections 1.4 and 7 carry +/- 2 % and are corrected to +/- 5 % by ECO against TST-EEG-004. This is a change to TST-EEG-004 and does not weaken the instrument specification; it stops the line from measuring something it cannot measure.

12.3 T8 noise

The RMS estimate over a 60 s record in a 69.5 Hz band has about 2 x 69.5 x 60 = 8340 degrees of freedom, so its relative standard deviation is 1/sqrt(2 x 8340) = **0.77 % of reading** (*calculated*) – negligible against the 1.0 uV limit. The real repeatability term is the fixture and the environment: mains pick-up into a shorted-input jig, and the operator's cable dress. That term has never been measured and must come out of the study, not out of arithmetic. A guard band applies: any channel reading above **0.85 uV** is re-tested with the unit inside a screened enclosure before it is failed.

12.4 Golden unit and fixture control

- Unit **TIOV-B-0001** is retained by the programme as the golden unit and is re-measured on the production fixtures at the start of every build lot. Where another document names the golden unit differently, this serial and the format of section 9 govern.
- Its T7 gains, T8 noise and T10 offsets are the fixture's control chart. **Gain drift > 0.2 % or noise drift > 0.15 uV halts the line** until the fixture is investigated.

- Fixture self-test at shift start: divider ratio against a 6.5-digit DMM, the three reference resistors (**4k99, 10k0 and 49k9**, the E96 parts FIX-01/A actually switches – JIG-EEG-009 Rev C section 0.1 and TST-EEG-004 Rev C T10 name the same three), continuity of all 12 J14 pins and the three DIN plugs. Logged. *Was: "5 k, 10 k, 50 k"* – a rounding of the fitted parts, and one that does not stay harmless at the top of the range: RISK-EEG-011 Rev B H-24 records that 49k9 read through 47 kOhm and the 10 nF shunt gives **92.5 kOhm**, which is not the reading a true 50 k reference would produce, so a self-test written to the rounded name checks the fixture against the wrong number. *Corrected 2026-09-02: the series value is now the **68 kOhm** of ECO-EEG-024, so the same point reads **111.5 kOhm** (TST-EEG-004 T10). The point about the rounded name is unchanged; the number it is made with has moved, and RISK-EEG-011 H-24 has not yet been restated.*
- Test equipment on a 12-month traceable calibration: DMM, function generator, current shunt, **47.0 Ohm acoustic load** matching the shipped ATH-M20x (A-04 restated as 32 to 64 Ohm, and the calibrated output level measured per model), artificial ear for the E-29 type test, **500 V DC insulation-resistance tester** for FH-05, leakage measuring device, RF receiver, and the FIX-01/E colorimeter head used for the contact-light step. **There is no per-unit hipot station**: the 2.5 kV RMS type test is the isolator supplier's certificate. Asset numbers are recorded in every unit's calibration record, so that a drifting instrument can be traced back to the units it touched.
- Operator qualification: IPC-A-610 class 2 certification for assembly operators; a named, trained operator for the provisioning and test station.

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13. First-delivery audit checklist

The programme performs this once, on the first delivery of each phase, at goods-in in Brussels, against the shipped units and the shipped records. It is a document and product audit, not a site audit; a site audit is not required for this programme.

#	Question	Evidence	Pass criterion
A-01	Is there an IQC record for every lot in the build?	IQC pack	One closed record per lot, none open
A-02	Are the ADS1299 date codes and lots recorded per unit?	DHR	Present for both modules on every unit (RFQ section 10)
A-03	Is the cell lot and its UN 38.3 report reference recorded per unit?	DHR	Present on every unit
A-04	Is the ADuM4160 isolation certificate on file?	Supplier pack	Present, >= 2.5 kV RMS, matching the module part number
A-05	Were the six DIN retention NPTH holes unplated?	Bare-board FAI, boards in hand	Bare laminate, no mask, on every board inspected

#	Question	Evidence	Pass criterion
A-06	Is the isolation keep-out clear of copper, solder and flux, on all four layers?	Photograph per board + inner-layer artwork + inspection of 2 units	Clear
A-07	Are R90 and R91 each fitted exactly once, with no wire bridge?	Inspection of 2 units	Confirmed
A-08	Is C1-C16 C0G, not X7R?	Reel photographs + DHR	GCM1885C1H103JA16D or an approved ECO
A-09	Does the FAI report exist and is it approved?	FAI Forms 1-3	Signed by the programme technical lead before the batch
A-10	Does every unit have a complete TST-EEG-004 record with actual values?	Calibration records	No blanks, no "pass" in a numeric field
A-11	Are failed attempts retained, not overwritten?	Records	Retest history visible
A-12	Is first-pass yield reported per test step?	Lot summary	Present
A-13	Do the label, the calibration record and the USB iSerial agree on 2 sampled units?	Read back on the bench	Identical serial in the `TIOV-B-nnnn` form, and identical fingerprint
A-14	Was the golden unit re-measured at the start of the lot?	Fixture log	Present, within the drift limits
A-15	Is the fixture self-test logged for every shift in the build?	Fixture log	No gaps
A-16	Are the test instruments in calibration, with asset numbers in the records?	Certificates	All in date
A-17	Is there an ESD log for every shift in the build?	ESD log	Present, EPA verified

#	Question	Evidence	Pass criterion
A-18	Were any deviations or concessions applied, and were they approved?	NCR / concession file	Every one carries a programme signature and an ECO number
A-19	Are open NCRs closed or dispositioned?	NCR register	None open against delivered units
A-20	Do the units match the released configuration?	As-built record vs the document register	BOM revision, Gerber revision, firmware hash all match
A-21	Does the kit content match the packing list, on 2 sampled kits?	Unpack	Every line present, foam cut-outs correct (M-06), calibration certificate in the lid pocket
A-22	Is the delivery free of a spare cell?	Unpack	No spare cell in the case (PKG-EEG-015 section 7)
A-23	Is there a micro-section, a registration report and a plane-continuity result for every bare-board lot in the build?	Fabricator pack + IQC pack	Present per lot; four layers in the right order, internal annular ring >= 0.025 mm, misregistration <= 0.125 mm

A finding at A-05, A-06, A-07, A-08 or A-23 stops acceptance of the whole delivery. Findings elsewhere are raised as corrective actions under section 8.

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14. Records, and what is still open

14.1 Records this plan creates

Record	Owner	Retention
IQC record per lot, including the four-layer coupon results	Manufacturer	7 years
FAI Forms 1-3 per phase	Manufacturer, approved by the programme	Life of the fleet + 10 years
In-process tick sheets, AOI reports, ESD log, reflow profiles	Manufacturer	7 years
TST-EEG-004 record and calibration record per serial	Both	Life of the fleet + 10 years (programme copy)
NCR, MRB minutes, concessions	Both	Life of the fleet + 10 years

Record	Owner	Retention
Corrective action reports	Both	7 years
Gauge R&R study, golden-unit control chart, fixture logs	Both	Life of the fleet
Device history record	Programme master	Life of the fleet + 10 years

14.2 Open items

Not yet decided or not yet possible. These are stated here rather than papered over.

1. **No safety engineer has reviewed this design.** The electrical safety review named in RFQ section 9.2 has no appointed reviewer. Rows FB-16, FA-10, A-04 and A-06 above are written so that the evidence exists when a reviewer is engaged, but no sign-off can be claimed until then. This blocks use on a person; it does not block fabrication or quoting. The move to four layers and the ECO-EEG-023 comparator change are both specifically in that reviewer's scope.
2. **No hardware has been manufactured or measured.** Every design value in Form 3 marked *calculated* – 48.77 Hz, Q 0.7416, 1.59 Hz, 52 mV, and, at the 68 k of ECO-EEG-024, 0.75 dB and 0.31 uV – is a calculation from design.py and has never been observed on hardware. *Corrected 2026-09-02: the loss and noise figures listed here were the 47 k pair, 0.36 dB and 0.27 uV.* The firmware image in `firmware/release/` is the one thing in the package that has been built, and it has run only under QEMU emulation. The FAI is the first occasion on which any of them becomes a measurement.
3. **S-02 is met in the design, and the sign-off is what is still open.** *Corrected 2026-09-02.* ECO-EEG-024 is applied in `tools/design.py`: R1-R16 are 68 kOhm, so the single-fault DC current through one protection resistor is **36.8 uA calculated against the 50 uA limit**, the corner moves to 234 Hz and E-10 sits at the +/- 1.0 dB branch it already carried. Rev B's "S-02 is not met as built … 53.2 uA … the Phase 1 prototypes are built at 47 kOhm" is superseded on that date; the resistor changed in the design before any prototype was built. **What is open**: the figure is calculated and no unit exists to measure it on, and the electrical safety reviewer of item 1 owns the disposition and has not started. S-02 stays a safety-reviewer item; it is no longer a stated non-conformance.
4. **The S-04 thermistor is not met and stays not met, and it takes half of E-23 with it.** There is no NTC net in design.py and no thermistor way on J12 or J13, so nothing on the carrier can read the cell temperature. **E-23's 45 C charge inhibit is therefore not met and cannot be tested either**; E-23's other half, thermal regulation inside the charger IC, is met on the module, and the two halves must always be stated together. The two hardware holes are the same hole. No closure is proposed. This is an open hardware item carried in DSN-EEG-003 section 11, RISK-EEG-011 and RFQ-EEG-001 Rev E section 12 item 3, and IQC cannot inspect its way round it.
5. **E-24 is not met.** The named isolator module presents a USB-B host receptacle where E-24 asks for USB-C. The interim answer is the WH-09 USB-B to USB-C panel pigtail, and it stays a live non-conformance until an isolator module with a USB-C host connector is qualified. The host connection is a socket in a gasketed aperture, not a captive cable; the captive lead through a gland is a Phase 2 item.

6. **The boom preamplifier part is not settled.** The MAX9814 is not approved because its AGC is forbidden by E-14; a fixed-gain part of the MAX4466 class is preferred; the module is specified by interface in ICD-EEG-006 until one is bought and measured. IQC row "boom preamplifier module" is written against that interface, not against a part number.
7. **The contact-light driver is written, and one of its states still cannot be reached.** *Corrected 2026-09-02.* Rev B's "`lights_write()` and `lights_task()` are on/off only; the bicolour phase scheme at 240 Hz is specified and not yet coded, so the contact-light test step cannot pass" is superseded: `firmware/main/main.c` captures both halves of the converter's lead-off status and derives a colour per site – trips neither threshold **green**, trips only the sensitive one **amber**, trips both **red** – alternating the two phases at about **250 Hz**, which is `LIGHT_PHASE_HZ` = 240 quantised to the FreeRTOS tick and meets E-27's "above 100 Hz". TST-EEG-004 T11 is written against it. **Corrected again 2026-09-02 (FW-D17):** this item read that the colour came from two detectors and that `LOFF_SENSN` was never written, so red could never be set. Red is reachable now, from a swept positive-side comparator threshold rather than from the N detector, which this single-ended montage does not have. **What still gates lot release**: T11 has never been run, because no unit exists; and the two `COMP_TH` settings are datasheet endpoints, so the impedance boundaries between green, amber and red are not yet established (TST-EEG-004 T11 Note 3). That is a firmware-owner item with E-27's owner, and no unit is released against T11 until the boundaries are set and T11 has actually been run.
8. **The gauge R&R numbers in section 12 are the estimator floor only.** The fixture and operator terms are unknown until the first study runs. If the measured %GRR at the 1 mV point exceeds 20 %, the fixture is the problem and the limits are not to be relaxed to make it pass.
9. **X-ray scope is open, and the two governing documents name different objects.** RFQ-EEG-001 section 9.1 asks for X-ray of the module connectors on the Phase 1 units; this plan and AVL-EEG-017 ask for X-ray of the converter packages. Neither appears as a numbered step in TST-EEG-004 Rev C. Both are Phase 1 only. The programme has not decided which object is inspected, whether it extends to the fleet, or whether a sampling plan replaces it. Bidders should price both.
10. **Conformal coating: decided, and the decision is no coating.** The board is not coated for Phases 1 and 2. It lives inside a gasketed enclosure, and coating a board with 30 connectors and a socketed DevKit costs more in masking than it buys. Nothing is applied and nothing is inspected. Revisited before Phase 3 if a unit returns with corrosion. Rev A of this plan recorded this as an open decision; it is not one.
11. **Panel fiducials remain the fabricator's.** The board itself now carries three fiducials (ECO-EEG-020, rows IQC-B14 and FB-18) and the vision-teach workaround is withdrawn. Panelisation is still the fabricator's choice (fabrication note 12), so panel-level fiducials on the rails are theirs to add, and the FAI records what was actually provided.
12. **eFuses are not burned on the Phase 1 prototypes.** Secure boot and flash encryption start at Phase 2 so that the firmware volunteer can iterate on unsigned images. The eFuse step is a Phase 2 gate and is not run on the two prototypes.
13. **The fabrication data is released for review, not for fabrication.** The Rev B DRC report records zero violations: all 145 nets fully connected, every net one connected copper island, both inner planes continuous under the analogue zone,

and the isolation strip free of copper on all four layers. Those are the three conditions ECO-EEG-016 section 3 sets for a fabrication-data release, and all three are met, so the data is released for review under RFQ-EEG-002A. Boards still cannot be ordered against it: the routing came from the programme’s own tools, no human layout engineer has reviewed it, and 169 of its connections close at the minimum conductor or the minimum gap rather than the preferred 0.25 mm width. Row IQC-B15 is the inspection point, and the Rev B report passes it.